Stochastic Thermodynamics of the Potts Model

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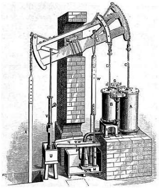
19th century thermodynamics:

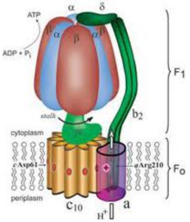


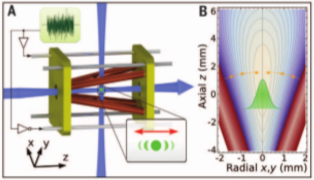
No dynamics

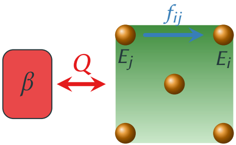
21st century thermodynamics:

$$\text{Small} \longrightarrow \begin{cases} \text{Fluctuations} \\ \text{Far-from-equilibrium} \\ \text{Quantum effects} \end{cases}$$

Build thermodynamics on dynamics







- System with states i of energy E_i and occupation probability p_i interacting with a heat bath at inverse temperature β and subjected to a nonconservative force f.

- Markovian master equation and local detailed balance

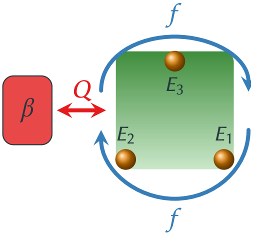
$$\dot{p}_i = \sum_j W_{ij}\, p_j\,, \qquad \ln \frac{W_{ij}}{W_{ji}} = -\beta\left(E_i - E_j - f_{ij}\right)$$

- First law of thermodynamics: $\mathrm{d}_t\langle E\rangle = \langle \dot{W}\rangle + \langle \dot{Q}\rangle$
  - Energy current $\mathrm{d}_t\langle E\rangle = \mathrm{d}_t \sum_i E_i\, p_i = \sum_{i,j} (E_i - E_j)\, W_{ij}\, p_j$
  - Work current $\langle \dot{W}\rangle = \sum_{i,j} f_{ij}\, W_{ij}\, p_j$
  - Heat current $\langle \dot{Q}\rangle = \sum_{i,j} (E_i - E_j - f_{ij})\, W_{ij}\, p_j = -\frac{1}{\beta}\sum_{i,j} W_{ij}\, p_j \ln\frac{W_{ij}}{W_{ji}}$

- Second law of thermodynamics: $\langle \dot{\Sigma}\rangle = \mathrm{d}_t\langle S\rangle - \langle \dot{S}_e\rangle \geq 0$
  - Nonequilibrium entropy $\langle S\rangle = -\sum_i p_i \ln p_i$
  - Entropy flow $\langle \dot{S}_e\rangle = -\sum_{i,j} W_{ij}\, p_j \ln\frac{W_{ij}}{W_{ji}} = \beta\langle \dot{Q}\rangle$
  - Entropy production $\langle \dot{\Sigma}\rangle = \sum_{i,j} W_{ij}\, p_j \ln\frac{W_{ij}p_j}{W_{ji}p_i} \geq 0$

- Detailed balance at equilibrium $f_{ij} = 0$: $W_{ij}p_j^{eq} = W_{ji}p_i^{eq} \Rightarrow \langle \dot{\Sigma}\rangle = 0$

Three-state unit with states 1, 2, 3 of energy E_1, E_2, E_3 and occupation probability p_1, p_2, p_3 interacting with a heat bath at inverse temperature β subjected to a nonconservative rotational forcing f.



- Markovian master equation and local detailed balance

$$\dot{p}_i = \sum_{j=1}^{3} W_{ij}\, p_j\,, \qquad \ln \frac{W_{ij}}{W_{ji}} = -\beta \left(E_i - E_j - f\,\Theta(i,j)\right), \qquad \Theta(i,j) = \begin{cases} 1 & (i-j) \mod 3 = 1 \\ -1 & \text{else} \end{cases}$$

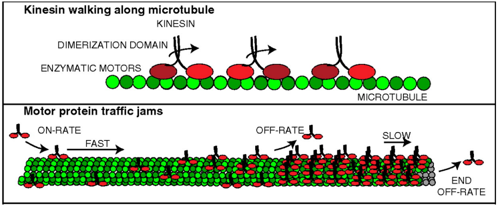
- Thermodynamics at steady state: $\dot{p}_i = 0$
  - $\mathrm{d}_t \langle E \rangle = \mathrm{d}_t \langle S \rangle = 0$
  - Nonequilibrium: $W_{ij} p_j \neq W_{ji} p_i$, $\langle \dot{\Sigma} \rangle \neq 0$
  - Clockwise current: $I \sim W_{32} W_{21} W_{13} - W_{31} W_{12} W_{23}$
  - $T \langle \dot{\Sigma} \rangle = \langle \dot{W} \rangle = -\langle \dot{Q} \rangle = 3fI$

- Detailed fluctuation theorems:

$$\textbf{Crooks} \quad \frac{P(W = w)}{P(W = -w)} = \mathrm{e}^{\beta w}$$

- Nonequilibrium phase transitions are most commonly discussed in the framework of nonlinear dynamics and bifurcation theory.
- Stochastic thermodynamics provides a systematic treatment of nonequilibrium thermodynamics $\Rightarrow$ thermodynamics of nonequilibrium phase transitions.
- Practical motivation to explore the thermodynamics of nonequilibrium phase transitions in large ensembles of interacting degrees of freedom:
  - Limitations of energy-converting nanomachines $\Rightarrow$ Consider ensembles of nanomachines
  - Phase transitions affect performance of large ensembles of machines?
  - Role of the number of nanomachines?
- Example:

Kinesin traffic jam

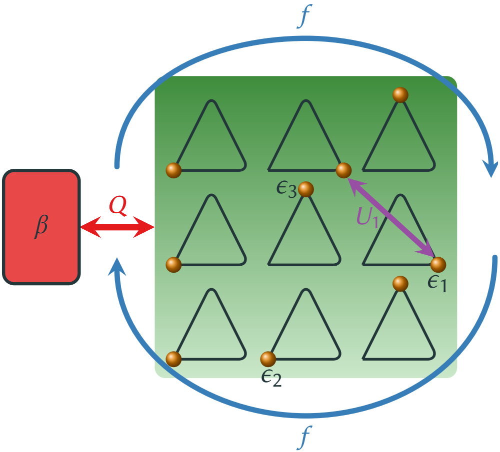


- Synchronization as a promising candidate for a beneficial performance of the ensemble.

$\Rightarrow$ Search for a (feasible) thermodynamic model exhibiting synchronization.

Part I: Three-State Potts Model

- Network of N all-to-all coupled three-state [PRL **96**, 145701 (2006)] units.
- Open system in contact with heat bath.
- Interaction energy $U(\mathbf{N}) = \frac{u}{2N}\sum_{k=1}^{3} N_k^2 + U_0$.
- Subjected to torque f acting as a global bias.



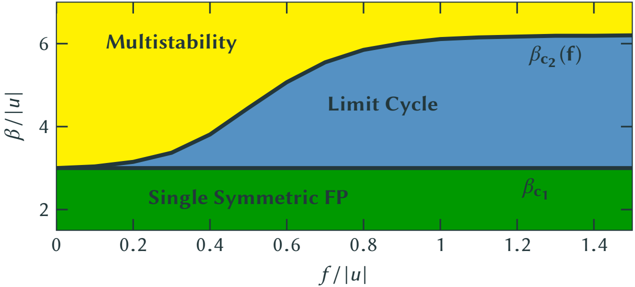
- Markovian master equation with Arrhenius rates satisfying local detailed balance

$$\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N},\mathbf{N}'}\, P_{\mathbf{N}'}\,, \qquad \ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}}} = -\beta\left[\Delta F^{eq}(\mathbf{N},\mathbf{N}') - \sigma(\mathbf{N},\mathbf{N}')f\right], \qquad F^{eq}(\mathbf{N}) = E(\mathbf{N}) - \beta^{-1}\overbrace{S^{int}(\mathbf{N})}^{\ln \frac{N!}{\prod_i N_i!}}$$

- Mean-field dynamics given by a nonlinear master equation

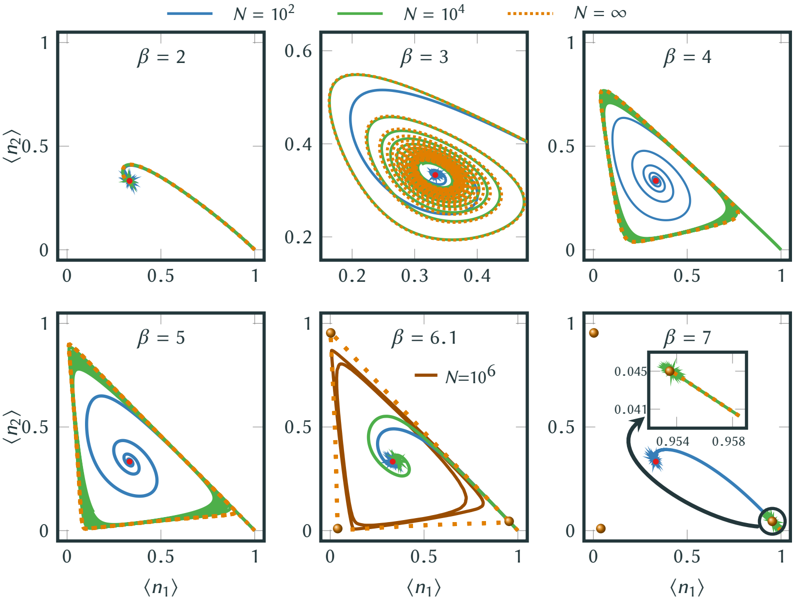
$$\dot{\overline{n}}_i(t) \equiv \langle \overline{n}_i(t)\rangle = \sum_{j=1}^{3} k_{ij}(\overline{n}_i,\overline{n}_j)\,\overline{n}_j(t), \qquad \ln\frac{k_{ij}}{k_{ji}} = -\beta\left[\epsilon_i - \epsilon_j + u(\overline{n}_i - \overline{n}_j) - \sigma(i,j)f\right].$$

- For $\epsilon_i = \epsilon$, the nonlinear equation has a symmetric fixed point $\dot{\bar{n}}_i = \sum_{j=1}^{3} k_{ij}\,\bar{n}_j\big|_{\bar{n}_j=1/3} = 0$.
- Linear stability indicates Hopf bifurcation: $\lambda_\pm = -\Gamma(\beta u + 3)\cosh\left(\frac{\beta f}{2}\right) \pm i\sqrt{3}\,\Gamma\sinh\left(\frac{\beta f}{2}\right)$.



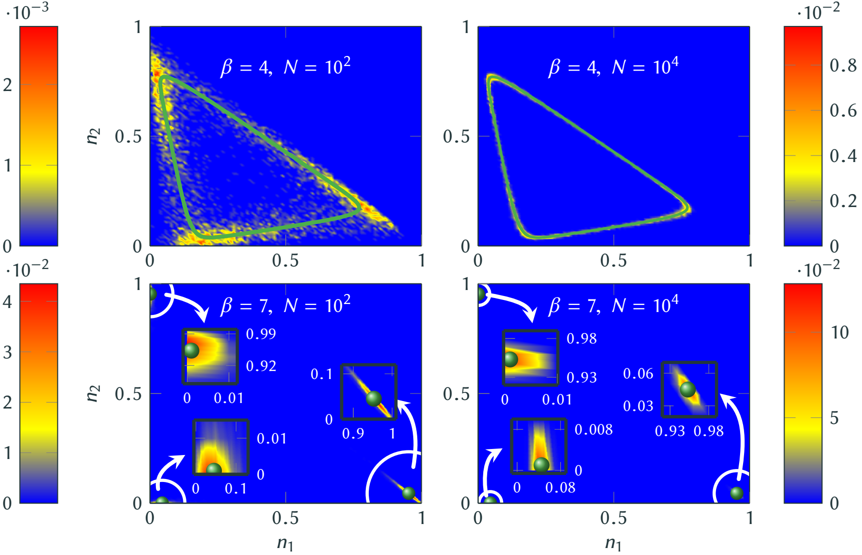
- One equilibrium (first-order) and two nonequilibrium phase transitions at $\beta = \beta_{c_{1,2}}$.
- Three distinct fixed points in the low-temperature regime ($\beta \gg \beta_c$) Energy-driven
- Symmetric solution in the high-temperature regime ($\beta \ll \beta_c$) Entropy-driven
- Thermodynamic consistency constraints stable oscillations to emerge only at intermediate temperatures ($\beta_{c_1} < \beta \leq \beta_{c_2}$).

$\Longrightarrow$ Minimal thermodynamic model exhibiting synchronization



$\Rightarrow$ Large system dynamics indeed agrees with the mean-field solution at metastable times.

Is metastability a generic property of our system or merely an artifact of special initial conditions?



$\beta = 4$

$N = 10^2$

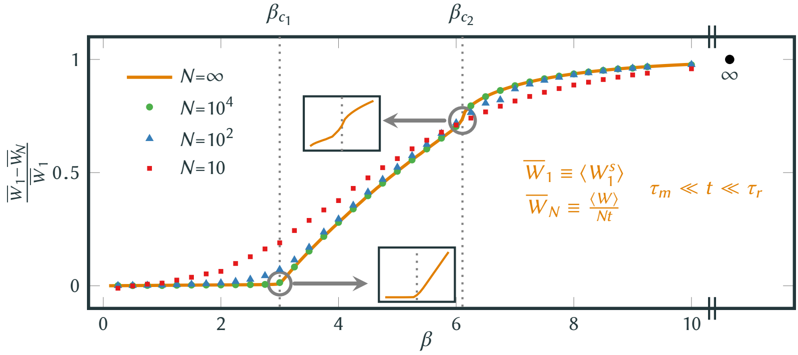
$N = 10^4$

$\beta = 7$

$N = 10^2$

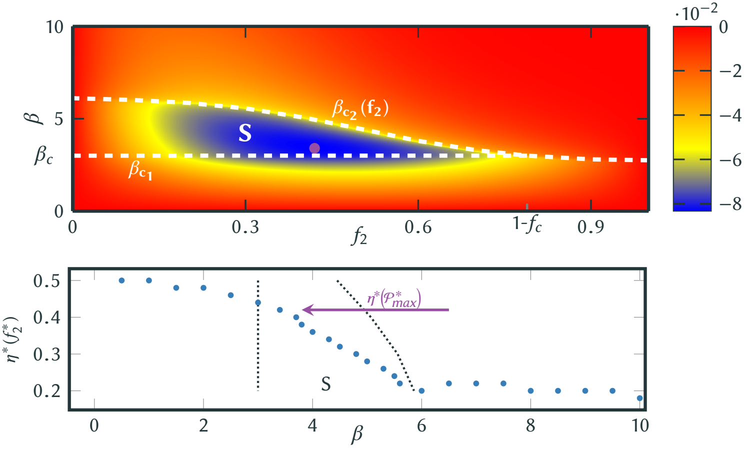
$N = 10^4$

- First Law: $d_t \langle E \rangle = \langle \dot{Q} \rangle + \langle \dot{W} \rangle$
- With this setup, work needs to be done on average to maintain the system, $\langle \dot{W} \rangle > 0$, which is on average dissipated as heat into the bath, $\langle \dot{Q} \rangle < 0$.



- (Signatures of) Nonequilibrium phase transitions at the critical points (for large but finite N). [“First-order phase transition” at β_{c_1} and “second-order phase transition” at β_{c_2}].
- Reduction of the dissipated work for finite interactions $u < 0$.

- Work-to-work conversion $f \equiv f_1 - f_2$, with input work $\mathcal{W}_1(f_1)$ and output work $\mathcal{W}_2(f_2)$.
- Maximization of the output power $\mathcal{P} \equiv \dot{\mathcal{W}}_2$ with respect to output force $\partial_{f_2} \mathcal{P}\big|_{f_1=1} = 0 \Rightarrow f_2^*$.
- Efficiency of work-to-work conversion at maximum power $\eta^* \equiv 1 - \frac{f}{f_1}\Big|_{f_2=f_2^*}$

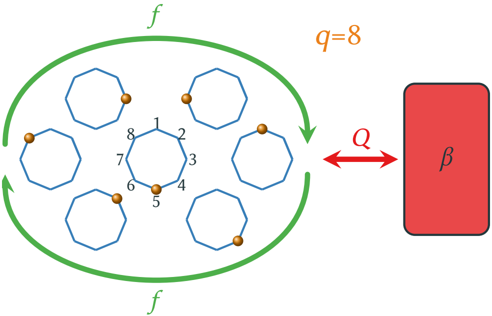


- Power output is maximized in the synchronization phase.
- The efficiency at maximum power, $\eta^*(f_2^*)$, is surprisingly close to the universal value, $\eta^* = 1/2$, for a tightly-coupled (single net-current) system responding linearly.

Part II: q-State Potts Model

q-State Unit Network

- Consider infinitely many globally interacting and driven q-state units on a ring.



- Stochastic dynamics is governed by a nonlinear Markovian master equation

$$\dot{\overline{n}}_i(t) = \sum_{j=1}^{q} k_{ij}(\overline{n}_i, \overline{n}_j)\,\overline{n}_j(t), \quad \ln\frac{k_{ij}}{k_{ji}} = -\beta\left[\epsilon_i - \epsilon_j + u(\overline{n}_i - \overline{n}_j) - \sigma(i,j)f\right].$$

- Flat energy landscape $\epsilon_i = \epsilon\ \forall i$ and attractive interactions $u < 0$:
- Symmetric solution $\overline{n}_i = 1/q$ in the high-temperature limit — Entropy-driven
- q energy-ground states $\overline{n}_i = \delta_{ij}$ in the low-temperature limit — Energy-driven
- Intermediate temperatures?

- Linear stability analysis of $\overline{n}_i^* \equiv 1/q$ yields for the critical point $q + \beta_c u = 0$ for $q \leq 4$.
- The linear stability matrix $\mathbf{A}_{ij} \equiv \sum_k [\partial(k_{ik}\overline{n}_k)/\partial\overline{n}_j]|_{\overline{\mathbf{n}}^*}$ evaluated at $\beta_c = -u/q$ reads

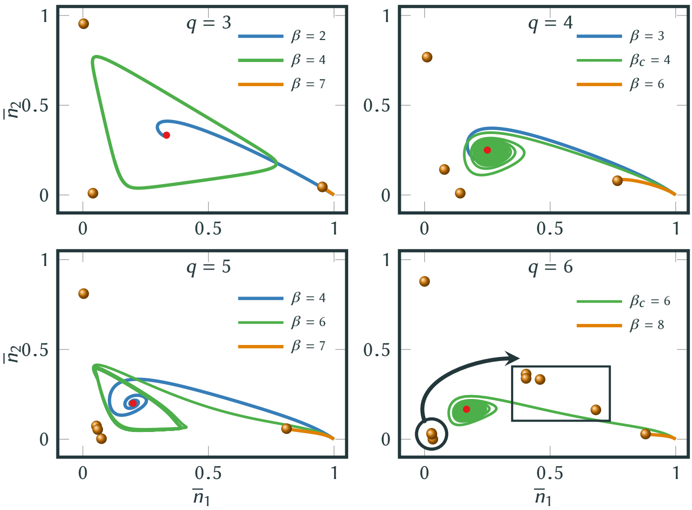
$$\mathbf{A} = \begin{pmatrix} 0 & c & 0 & \dots & 0 & 0 & -c \\ -c & 0 & c & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -c & 0 & c \\ c & 0 & 0 & \dots & 0 & -c & 0 \end{pmatrix}_{q\times q} \qquad \begin{aligned} c &= \Gamma \sinh\left(q\,\frac{f}{2u}\right) \\ \lambda_k &= 2\mathrm{i}\,c\sin\left(k\frac{2\pi}{q}\right), \quad k = 0, 1, \dots, q-1. \end{aligned}$$

- Perturbation of $\mathbf{A}(\beta_c + \delta\beta) = \mathbf{A}(\beta)\big|_{\beta_c} + \delta\beta\;\mathbf{B} + \mathcal{O}\big(\delta\beta^2\big)$, yields

$$\mathbf{B} = \begin{pmatrix} a & b & 0 & \dots & 0 & 0 & c \\ c & a & b & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & c & a & b \\ b & 0 & 0 & \dots & 0 & c & a \end{pmatrix}_{q\times q} \qquad \begin{aligned} & a = -\tfrac{2\Gamma u}{q}d, \;\; b = \Gamma\tfrac{2u-qf}{2q}d, \\ & c = \Gamma\tfrac{qf+2u}{2q}d, \;\; d = \cosh\left(q\tfrac{f}{2u}\right), \\ & \tilde{\lambda}_k = -\frac{2\Gamma d\sin^2\left(q\frac{k\pi}{u}\right)}{q}\left[2u + \mathrm{i}fq\cot\left(\tfrac{k\pi}{q}\right)\right], \; k = 0, 1, \dots, q-1. \end{aligned}$$

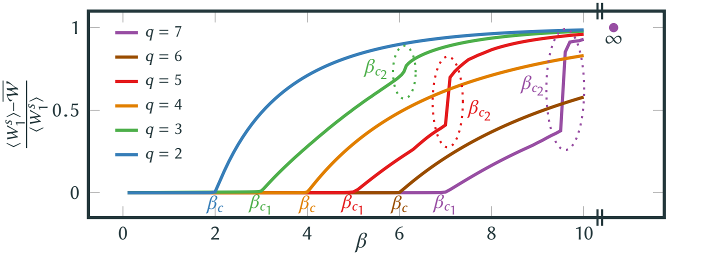
- In absence of driving, $f = 0$, the real eigenvalues change sign $\longrightarrow$ saddle-node bifurcation
- In presence of driving, $f \neq 0$, the real part of the eigenvalues change sign $\longrightarrow$ Hopf bifurcation

- Stability of oscillations?

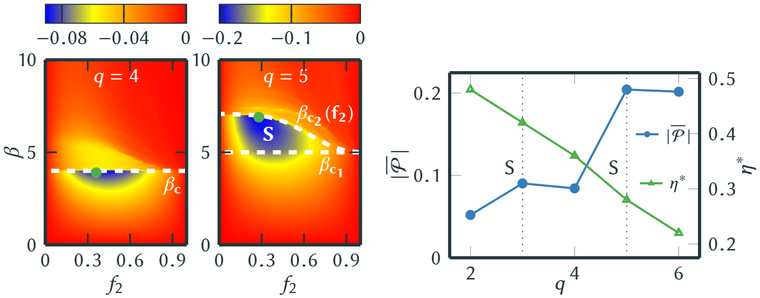


- If q is even there is only a subcritical Hopf bifurcation, if q is odd there are two bifurcations: a supercritical Hopf and an infinite-period bifurcation.

- Dissipated work



- Power-efficiency trade-off



Part I: Three-State Potts Model

- Minimal thermodynamic model exhibiting synchronization.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- The maximum output is achieved for a synchronous operating mode and the associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

Herpich, Thingna, and Esposito, PRX **8**, 031056 (2018)

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Part II: q-State Potts Model

- Two classes of Potts models (even vs. odd q) with different phenomenology.
- Small and odd-q Potts models (efficiency and synchronization) are energetically beneficial.

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Thank you for your attention!